

1. Overview

OncoGemma is a copilot designed for pathologists to help medical institutions provide same-day cancer diagnosis. This version of OncoGemma is designed to assist pathologists in handling H&E slides for breast tissue and generating a CAP-compliant report for breast cancer examination.

2. Observations

1. Pathology models developed by Google using VLM architecture are now very close on benchmarks to SoTA models developed at apex medical research facilities (Mayo)
2. Health AI models for cancer diagnostics are now available on usage-based pricing, reducing adoption costs.

3. Hypothesis

We believe the copilot can serve as a diagnostic tool for pathologists in cancer diagnostics. In this version, we want to build the complete pipeline for H&E examination of breast cancer (including all the steps as used by apex centres using AI models) using Google models and ML scripts. The copilot will take WSI slides as input, ask pathologists to confirm the intermediate steps and generate CAP-compliant reports as output.

4. Past Iterations

v1 (Feb '26)-  Code v1

We learned the different applications available with GenAI. We examined applications on MRIs, X-Rays, pathology, radiology. We noted the following:

1. Pathology models are the least susceptible to changes in human physiology, and resulting applications will be global in nature.
2. Cancer diagnostics as a space is really exploding with too much demand and diagnostic tests arriving in 5-10 days at the best centres in India while apex global centres deliver same-day diagnosis.
3. Breast cancer is the most common cancer. This results in easier validation and confirmation because of large public datasets

v2 (April '26)- <https://github.com/preshit-gupta/OncoGemma-v2>

We created a 1-shot pipeline for multiple breast cancer examinations to test the models and their outputs. We realised that the model sometimes hallucinates and provides garbage outputs from bad inputs. Therefore, we started working towards a copilot version where intermediate steps can be verified by the user.

v3 (June '26)- <https://github.com/preshit-gupta/OncoGemma-v3>

We created a copilot where pathologists can check and validate each step and confirm the outputs. We realised the rough edges of the medgemma models. We also had to include separate models to count mitotic figures. We also noted the need for independent and separate workflows for each cancer examination.

5. Workflow

Stage 1: Upload

Frontend- The user uploads whole slide images (WSI) of H&E examination for breast tissue. Currently, users can upload only 1 slide for analysis.

Backend- Unsigned GCP bucket to handle large WSI files

Stage 2: Slide Preprocessing

Process: Raw Whole-Slide Images (WSIs) are parsed and converted to the open-source, interoperable DICOM standard. To correct for chemical staining variations across different laboratories or scanner sensors, the pipeline parses the slide's embedded ICC color profile and runs the STAINS tool (Aequi) to normalize Hematoxylin and Eosin chromaticity.

Frontend- The user can view the slide for preview over a digital slide viewer that displays the slide after processing. The user continues to the next step after reviewing the slide.

Stage 3: Triaging Hotspots

Action: Screen out low-risk tissue and isolate the active invasive tumor front.

Process: The slide is downsampled to 10× magnification ($\sim 1.0 \mu\text{m}/\text{pixel}$) and segmented into patches. Patches are sent directly to Vertex AI Path Foundation model endpoint on Google Cloud.

Backend- We systematically create hotspots at 10x magnification for the entire slide. Each hotspot is passed through the Path Foundation model. This filters out empty areas and defines high-activity tumor hotspots.

Frontend- The user is shown the hotspots over an interactive digital slide viewer that displays composite dynamic SVG heatmaps. Pathologists can review and change these hotspots to refine the Regions of Interest (ROI). The user continues to the next step after reviewing the ROI.

Database- We store the cartesian coordinates, image patch and classification for every hotspot.

Stage 4: HPF Selection

Action: Rapidly locate and precisely verify cell-level biomarkers at 40× magnification ($\sim 0.25 \mu\text{m}/\text{pixel}$) restricted only to the hotspot coordinates mapped in Step 3. Implement a digitized mitotic counting process that mirrors classical microscopy through algorithmic spatial grids.

Process:

1. First-Pass Sweep (YOLO): The YOLO object-detection architecture sweeps the high-magnification hotspot patches to instantly flag coordinates of any dark, dense, hyperchromatic nuclear structures (candidate mitotic figures), successfully filtering out 99.9% of normal nuclei.

2. Second-Pass Verification (HoVer-Net): The candidate coordinates flagged by YOLO are cropped and passed to HoVer-Net. HoVer-Net performs precise instance segmentation and identifies true mitotic figures.
3. The pipeline generates a density heatmap using confirmed HoVer-Net coordinates. A circular kernel is applied with a radius equivalent to a standard 250–270 μm high-power field (HPF).
4. The system identifies the primary hotspot with peak mitotic density, automatically projecting a contiguous grid of 10 non-overlapping virtual circular fields across the tumor tissue.
5. Cumulative mitotic figures across these 10 HPFs are calculated to determine the Nottingham Mitotic Score (Score 1: 0–9; Score 2: 10–19; Score 3: ≥ 20)..

Frontend- The user is shown the HPF sites over an interactive digital slide viewer. They can review and change HPF areas. The user continues to the next step after reviewing the HPF sites.

Database- We store the cartesian coordinates of HPF sites, image patch, classification and cumulative mitotic figure count for every HPF site

Stage 5: Nottingham Grading and Synthesis

Action: Utilize MedGemma to evaluate Nottingham parameters and establish the final histologic grade.

Process:

1. High-activity tumor bed patches, preprocessed at 10 $\times$ magnification, are ingested by the MedGemma 1.5 architecture.
2. The MedSigLIP vision encoder conducts an automated assessment of architectural characteristics:
 - Tubule Formation: Grading is based on glandular structure proportions (Score 1: $>75\%$; Score 2: 10%–75%; Score 3: $<10\%$).
 - Nuclear Pleomorphism: Analysis of chromatin texture and atypia (Score 1: uniform; Score 2: moderate; Score 3: severe/vesicular).
3. The instruction-tuned text decoder integrates native visual findings with structured mitotic counts to generate the overall Nottingham Histologic Grade.

Frontend- The user is provided a summary of findings. The user continues to the next step after reviewing summarised findings.

Database- We store the patch classifications with counts, findings and nottingham grading

Stage 6: CAP-Compliant Reporting

1. Action: Consolidate diagnostic findings and prognostic markers into a standardized, audit-ready clinical dossier.
2. Process: MedGemma 1.5 synthesizes Nottingham parameters and tumor dimensions into structured element-response pairs. The output follows CAP protocols and includes a

natural-language summary. Integration of html2pdf.js to convert the dashboard into a high-resolution, print-ready clinical dossier.

6. Validation and Success Criteria

We are working towards a diagnostically accurate Software-as-a-Medical-Device (SaMD) that increases automation for highly specialised tasks and provides a user-friendly workflow for pathologists. If successful, this will lay the foundation of SaMD for cancer diagnostics in India.

Validation Methods (in order of ease):

1. Provide OncoGemma to 3-5 pathologists and oncologists for testing
2. Retrospective Validation Study on existing clinical data
3. Performance on existing benchmarks
4. External validation studies with institutional partners

7. Out of Scope

The following parts, while necessary can be added later when the basic workings of the product are validated:

- Ability to upload multiple slides of a single patient
- sTILS Density

8. Resources

H&E Examination (Pathologist)

<https://notebook.google.com/notebook/d85d29d6-93f5-4c4c-8ec5-4722f2d820bc>

H&E Examination (Computational)

<https://notebook.google.com/notebook/12106382-5d0c-4f9e-a77d-5e743a4797ee>

Suggested Workflow

<https://notebook.google.com/notebook/e53fc74f-8180-41bc-a19e-c4fbd22538d3>

Case Study: Evolution of CBC Testing

<https://notebook.google.com/notebook/6345a5a8-0aba-4d8c-91ca-b053bf0833bd>

MedGemma: Technical Report

[\[2604.05081\] MedGemma 1.5 Technical Report](#)